ASCII63, with its 7-bit *code unit* (the number of bits used to store a character value), meant that a whopping 128 codepoints were available in the code page. This was a previously unheard-of luxury; older systems might only have used six bits (looking at you, IBM of the 1950s), meaning only 64 possible codepoints. 7-bit ASCII meant both uppercase *and* lowercase Latin characters were possible (gasp!) with enough room left over for the Arabic digits, common punctuation, a smattering of currency symbols, and some control characters. Another aspect of seven-bit ASCII that ensured its success was the fact that the characters were *collatable*: rules existed within ASCII for comparing and sorting subsets of codepoints ("A" has a lower codepoint value than "B," both are lower than "a," and so on), and this was not always true of encoding systems, even those designed for computing, such as IBM's legendarily awful punchcard-derived *EBCDIC*. All of this is fantastic… right up until you need to use, say, Cyrillic characters.

"Extended ASCII"

To be fair, the ASCII committee likely didn't have the needs of Russian (or any other non-English) users uppermost in their minds in the 1960s. Other standards quickly emerged to meet the growing needs of the rest of the world, beginning with ISO/IEC 646 (ECMA-6), which was another 7-bit system closely related to ASCII but allowed variation of certain codepoints; multiple national standards were derived from it. Next came probably the most significant, ISO-8859 (offering 256 codepoints at eight bits per point), which consisted of a series of standards for encoding multiple languages. Examples include ISO-8859-1 (Latin-1) for Western Europe and ISO-8859-5 (Cyrillic) for Serbia, Ukraine et al.

These multiple 8-bit encoding schemes are usually collectively known as *extended ASCII*, but that is something of a misnomer: the ANSI had nothing to do with their creation. Nevertheless, a stop-gap solution had been found to cram a lot more characters into existing systems, and all of this too was fantastic… right up until you need to, say, open a text file from a Serbian ISO-8859-5 system on one using ISO-8859-1. Oh, and for extra giggles, toss in some proprietary "extended" sets such as those developed for DOS and Windows. Hopefully the inherent problems of multiple overlapping standards are obvious (assuming you didn't experience it firsthand like some of us had too): in an era when systems from very different countries and cultures were increasingly interacting with each other, this approach to character encoding was throwing up problems with information transfer.